Zero-shot language models versus accumulating local data for timeline-aligned prediction in paediatric acute lymphoblastic leukaemia

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Abstract

Prediction models for paediatric acute lymphoblastic leukaemia (ALL) are usually trained on a hospital's own historical cases, yet a hospital that adopts a new protocol starts with none. Here we ask how much local data a zero-shot large language model (LLM) is worth along the clinical timeline. We aligned seven nested feature sets with data acquisition from admission to Day 46, defined six decisions—Final Risk, day-19 and day-46 minimal residual disease (MRD), relapse, event-free survival (EFS) and overall survival (OS)—and trained random forests and random survival forests on cases partitioned by enrolment year (301 patients, 2015–2019), with an external cohort of 41 patients from two hospitals. An equal-weight ensemble of four zero-shot LLMs matched models trained on about 90–120 local patients for day-19 MRD and EFS, and outperformed locally trained random survival forests for OS in every evaluable test set (C-index 0.71–0.83 versus 0.61–0.64). Trained models were superior only for Final Risk, which a three-condition protocol rule reproduced in 99.3% of patients and which they learned from about 30 cases. At the external hospitals the trained EFS model lost discrimination (C-index 0.75 to 0.61) whereas the LLM ensemble changed little (0.71 to 0.67). Relapse and day-46 MRD remained near chance for all methods. Zero-shot LLMs are therefore a credible starting point before local data accrue, and remain complementary afterwards.

Keywords: paediatric acute lymphoblastic leukaemia, large language models, zero-shot prediction, learning curves, temporal validation, survival analysis, external validation

1 Introduction

Acute lymphoblastic leukaemia (ALL) is the most common childhood malignancy, accounting for approximately 25% of all paediatric cancers [1]. Risk-adapted chemotherapy has elevated five-year event-free survival (EFS) rates above 90% in high-income settings, yet the 15–20% of patients who relapse face substantially worse outcomes, with long-term survival dropping to 30–50% upon re-induction [2, 3]. Because the majority of relapses occur after apparent clinical remission, the prospective identification of high-risk patients—early enough to intensify or de-escalate therapy—remains a central and unresolved challenge in paediatric oncology.

Current standard-of-care risk stratification—Initial Risk at diagnosis and Final Risk after the Day 46 induction assessment—relies on pre-specified categorical cut-offs for age, white blood cell count, cytogenetics, immunophenotype and minimal residual disease (MRD), established in large cooperative-group studies [4–6]. These rules do not yield personalised probability estimates for future relapse or survival, and a growing literature has sought to provide them: Cox-regression nomograms [7–10], risk scores and association studies built on the day-19 and day-46 MRD time points of the CCCG-ALL-2015 protocol [11, 12], machine-learning models on routine clinical variables [13–17], and models built on gene-expression, single-cell, exome, methylation or RNA-sequencing profiles [18–23]. Relapse has proved hardest: computational flow cytometry at diagnosis failed to predict it [24], and the largest machine-learning relapse model reached 68.5% sensitivity at 50.3% specificity [17].

Most of these models make simplifications that break down at the bedside (Table 1). They typically predict one endpoint at one decision point, and they are validated by random split or cross-validation: a systematic review found external validation in only 5 of 20 machine-learning models for paediatric haematological malignancies [25], a meta-analysis of childhood-cancer prognosis models reported that no study had yet assessed AI-based models in haematological malignancies [26], and none of the paediatric ALL studies we identified used a temporal split by enrolment year. The work closest to ours, clinTALL [27], jointly predicts T-ALL subtypes, competing events and EFS with a multi-task network restricted to pre-induction variables, but it does not organise features along the acquisition timeline and did not quantitatively evaluate outcome prediction on its external cohorts.

We argue that realistic clinical prediction is governed by three distinct and largely independent timelines, and that existing paediatric ALL models address them at most partially. The first is the *feature-acquisition timeline*: a patient's data do not arrive at once but in a fixed clinical order—white blood cell count and age at admission (Day 0), immunophenotype (Day 1), fusion genes, karyotype and FISH (Day 7–10), and MRD (Day 19–46). A model trained on features pooled across all timepoints implicitly uses information unavailable at the moment of decision, producing optimistic but non-deployable estimates [28]. The second is the *prediction-target timeline*: clinicians do not make one prediction but a sequence of decisions—early response (D19MRD), end-of-induction response (D46MRD), Final Risk assignment, and long-horizon outcomes (relapse, EFS, overall survival [OS]). Studying any single target in isolation, as is common, obscures how one data-acquisition workflow must support all six decision nodes. The third is the *data-accumulation timeline*: a hospital's reference database

grows year by year, so a model deployed in a given year can only have learned from patients enrolled before it. Random train–test splits ignore this constraint, so they cannot show how a model behaves as a hospital's database grows.

The third timeline exposes a practical problem that the first two do not: a hospital that adopts a new protocol, or a centre that begins to collect structured data, starts with no local cases to learn from. Large language models (LLMs) can be queried zero-shot, drawing on clinical knowledge acquired during pretraining [29]. On adult hospital data, language models have predicted readmission, mortality and other outcomes after domain pretraining, with temporal and prospective validation [30], or zero-shot from tokenised patient timelines [31], and LLM-extracted features from clinical notes have improved survival models in adult cancer [32]. With serialised structured data, however, the value of prompted LLMs depends on the comparator: they have often outperformed conventional models trained on ten examples [33] yet fallen well short of fully trained ones [34, 35]. Retrieval of similar patients or of external knowledge can help [36–40], but none of these studies concerns paediatric oncology. It therefore remains unclear how much local data a zero-shot LLM is worth, for which decisions it suffices, and whether it remains useful once a case database has accrued or when a model is moved to another hospital. We address these questions on a dataset in which seven temporally ordered clinical feature sets (six strictly nested; $\mathcal{F}_4$ is a Day-1 branch of $\mathcal{F}_3$) map onto the feature-acquisition timeline, six prediction targets map onto the target timeline, and enrolment year defines the data-accumulation timeline. We compare an ensemble of four zero-shot LLMs (Phase 1) with decision trees, random forests and random survival forests trained on the growing case database under temporal splits (Phase 2), test both on an external cohort from two hospitals without retraining, and add a retrieval-based belief revision that uses the accumulated cases to revise model probabilities and to ground LLM-generated clinical narratives.

Our contributions are threefold. (*i*) A timeline-aligned evaluation of six clinical decisions in paediatric ALL that shows when each becomes predictable: Final Risk follows a protocol rule, EFS discrimination emerges only with Day-46 MRD, and relapse and day-46 MRD remain near chance with routine data. (*ii*) A quantification of the value of zero-shot LLMs relative to accumulating local data: the LLM ensemble matched models trained on about 90–120 local patients for day-19 MRD and EFS, outperformed locally trained survival forests for OS in every evaluable test set, and lost less discrimination at external hospitals, whereas trained models were superior for the protocol-defined Final Risk. (*iii*) A retrieval-based belief revision extended to censored survival outcomes, with LLM-generated narratives for physician review.

2 Results

2.1 Cohorts and timeline-aligned design

The internal cohort comprised 301 children with ALL treated at [HOSPITAL NAME] between 2015 and 2019 (Table 2). Prediction targets span the treatment timeline: Day 19 MRD positivity (D19 MRD $\geq$ 1%), Day 46 MRD positivity (D46 MRD $\geq$ 0.01%), Final Risk (LR versus IR/HR; 159 LR, 142 IR/HR), relapse (44 of 301, 14.6%), event-free survival (EFS; relapse, induction failure or death; 75 events, 24.9%) and

Table 1 How this study differs from prior outcome-prediction work. Families of prior studies compared on the dimensions addressed here; individual studies are characterised in Supplementary Table S1. ✓, addressed; (✓), addressed partly or by some studies in the family; –, not addressed. Time-ordered features: predictors organised by the time at which they become available. Classification + survival: classification and time-to-event targets evaluated on the same features. Temporal split: training and test sets separated by calendar time. External cohort: outcome prediction evaluated on a cohort from other institutions. Learning curve: performance compared with models trained on different amounts of local data. Retrieval: retrieval of similar cases or external knowledge.

Family of prior work	Paediatric ALL	Time-ordered features	Classif. + survival	Temporal split	External cohort	Zero-shot LLM	Learning curve	Retrieval
Protocol risk rules and MRD studies [4–6]	✓	(✓)	–	–	–	–	–	–
Cox nomograms and risk scores [7–12]	✓	–	–	–	–	–	–	–
Machine learning on routine clinical data [13–17]	✓	–	–	–	–	–	–	–
Omics, single-cell and cytometry models [18–24]	✓	–	(✓)	–	(✓)	–	–	–
Multi-task outcome prediction (clinTALL) [27]	✓	–	✓	–	–	–	–	–
Language models on adult health records [30–32]	–	–	–	(✓)	–	(✓)	–	–
Prompted LLMs versus conventional models [33–35]	–	–	–	–	–	✓	(✓)	(✓)
Retrieval-augmented clinical prediction [36–40]	–	–	–	(✓)	–	(✓)	–	✓
This work	✓	✓	✓	✓	✓	✓	✓	✓

overall survival (OS; 24 deaths, 8.0%). The external cohort pooled 41 patients from two hospitals: Lianyungang First People's Hospital (20 patients, diagnosed 2019–2023) and Children's Hospital of Fudan University, Shanghai (21 patients, 2017–2021). The Fudan patients form a relapse-enriched referral series in which every patient relapsed, so the pooled external relapse rate (23 of 41) is far above that of an unselected ALL population. No external data were used for training.

Figure 1 summarises the design. Seven feature levels $\mathcal{F}_1$–$\mathcal{F}_7$ follow the order in which data become available, from white blood cell count and age at admission to D19 and D46 MRD at the end of induction (variable list in Supplementary Table S3). Each target is predicted at a pre-specified decision level, the latest level available before the corresponding clinical decision: $\mathcal{F}_6$ for D19 and D46 MRD, and $\mathcal{F}_7$ for Final Risk, relapse, EFS and OS. At each level we compare an equal-weight ensemble of four zero-shot LLMs (Gemini-3.1-Flash-Lite, Gemma-4-31B-it, InternVL3.5-30B-A3B and Qwen3.6-35B-A3B), which uses no training data, with random forests (RF) and random survival forests (RSF) trained on patients from earlier enrolment years. The primary internal test set is the 2018–19 enrolment cohort (Split 2; 186 training and 114 test patients); the other two temporal splits are reported in Supplementary Table S4. Discrimination is summarised by AUROC for classification, where the LLM

ensemble's score is the fraction of the four models predicting the positive class, and by Harrell's C-index for survival.

Table 2 Cohort characteristics and prediction target distributions. The External cohort pools patients from two independent hospitals (see Methods for site details). † patients with complete labels for all four classification targets. MRD percentages are of patients with a recorded value (internal D19MRD $n = 299$, D46MRD $n = 298$; external $n = 29$ and $n = 31$).

Characteristic	**Internal** ($n = 301$)	**External** ($n = 41$)
Enrolment years	2015–2019	2017–2023
B-ALL	263 (87.4%)	38 (92.7%)
Patients with complete labels†	298	29
Classification targets		
Final Risk: LR	159 (52.8%)	17 (41.5%)
Final Risk: IR/HR	142 (47.2%)	24 (58.5%)
Relapse: Yes	44 (14.6%)	23 (56.1%)
D19MRD $\geq 1\%$	42 (14.0%)	4
D46MRD $\geq 0.01\%$	29 (9.7%)	1
Survival endpoints		
EFS events	75 (24.9%)	24 (58.5%)
OS events	24 (8.0%)	7 (17.1%)

2.2 What becomes predictable, and when

Figure 2 shows discrimination for every target at every feature level, in the internal test set and at the external hospitals. Three patterns stand out.

Final Risk is largely determined before it is assigned. Trained RF reached an AUROC of 0.87 from white blood cell count and age alone and 1.00 at $\mathcal{F}_7$; the LLM ensemble rose from 0.79 to 0.95. This follows from how Final Risk is defined. In our cohort a three-condition rule—Initial Risk of IR or HR, D19 MRD $\geq 1\%$, or D46 MRD $\geq 0.01\%$—reproduced Final Risk in 299 of 301 internal patients (99.3%) and in 40 of 41 external patients, and Initial Risk alone reproduced it in 90.7%. Final Risk at $\mathcal{F}_7$ therefore tests whether a model recovers the protocol rule rather than whether it predicts an unknown outcome. The clinically uncertain part is which of the 187 initially low-risk patients will be upgraded (28 patients, 26 of them MRD-positive), which is the MRD prediction problem.

Survival discrimination appears late. The RSF C-index for EFS stayed between 0.53 and 0.61 up to $\mathcal{F}_6$ and rose to 0.75 only when D19 and D46 MRD entered at $\mathcal{F}_7$, consistent with the central role of MRD in risk stratification [5, 6]; the LLM ensemble rose from 0.68 to 0.71 over the same step. For OS, the LLM ensemble reached 0.70–0.74 from Day 5 onwards, whereas the RSF never exceeded 0.63.

Relapse and D46 MRD remain close to chance. Neither approach exceeded an AUROC of 0.67 for either target at any level. D19 MRD, by contrast, became

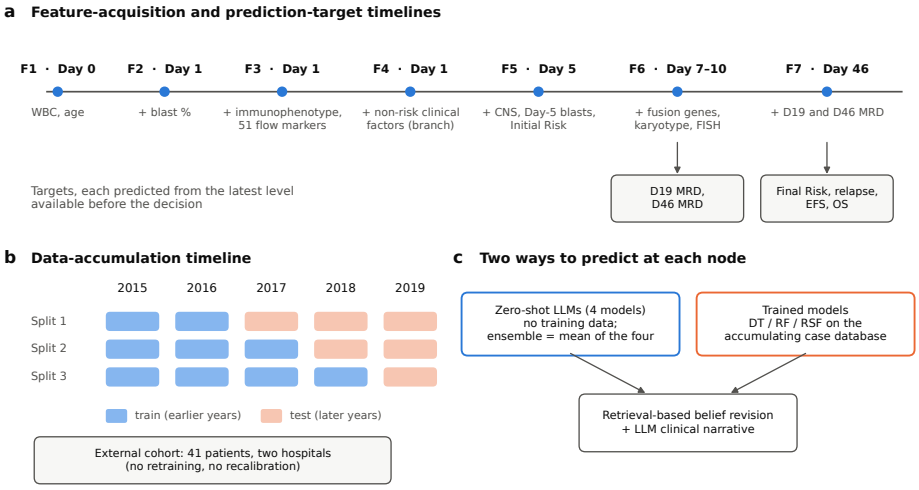


Fig. 1 Study design along three timelines. a, Seven nested feature levels follow the clinical order of data acquisition from admission (Day 0) to the end of induction (Day 46); each target is predicted from the latest level available before the corresponding decision. **b,** Trained models learn from earlier enrolment years and are tested on later years (three temporal splits) and, without retraining, on an external cohort from two hospitals. **c,** At each level, predictions come either from four zero-shot LLMs, combined by equal-weight averaging, or from decision trees, random forests and random survival forests trained on the accumulating case database; retrieval of similar past cases revises the trained models' probabilities and grounds an LLM-generated clinical narrative.

predictable once immunophenotype was available (RF 0.42 and 0.50 at $\mathcal{F}_1$ and $\mathcal{F}_2$, 0.74–0.81 from $\mathcal{F}_3$; LLM ensemble 0.78 from $\mathcal{F}_5$). Results for each LLM and each level are given in Supplementary Tables S5–S8.

2.3 How many local patients is a zero-shot LLM worth?

To quantify the value of zero-shot LLMs relative to local data, we retrained RF and RSF on random subsets of m patients from the 2015–17 training years and evaluated them on the same 2018–19 test patients as the LLM ensemble (Fig. 3; 20 repeats per size). The LLM ensemble matched trained models built from about 33 patients for Final Risk, 93 for D19 MRD and 120 for EFS. For OS, the RSF stayed below the LLM ensemble at every training size, including all 186 training patients.

With all training data (Table 3), trained models were better only for Final Risk (AUROC difference −0.050, 95% CI −0.097 to −0.010). They were statistically indistinguishable from the LLM ensemble for D19 MRD, D46 MRD, relapse and EFS, and worse for OS (C-index 0.737 for the LLM ensemble versus 0.629; difference +0.109, 0.011 to 0.229). The OS result held in the earlier split trained on 2015–16 (difference +0.105, 0.024 to 0.190). These OS training sets contained only 8 and 13 deaths, so a model that must learn from local events has little to learn from, whereas the LLMs bring prognostic knowledge from pretraining.

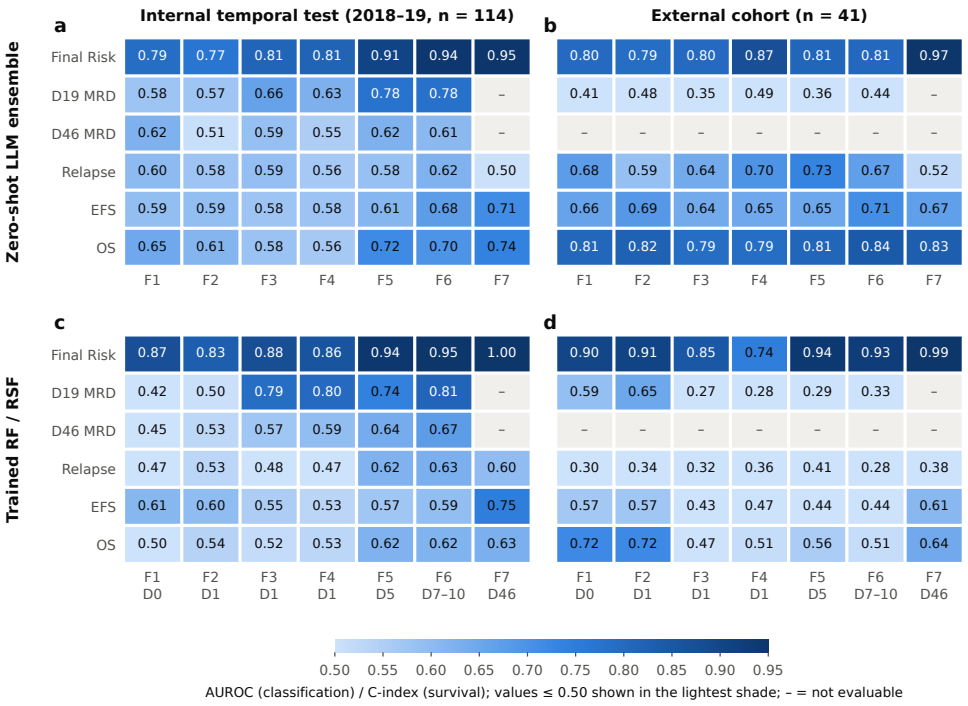


Fig. 2 What becomes predictable, and when. Discrimination of each target at each feature level ($\mathcal{F}_1$–$\mathcal{F}_7$, with the day on which the level becomes available) for the zero-shot LLM ensemble (**a**, **b**) and for trained random forests (classification) or random survival forests (survival) (**c**, **d**), in the internal 2018–19 test set (**a**, **c**; trained on 2015–17) and in the external cohort (**b**, **d**; trained on all internal patients). Values are AUROC for Final Risk, D19 MRD, D46 MRD and relapse, and Harrell's C-index for EFS and OS. –, not evaluable (MRD values are inputs at $\mathcal{F}_7$; only one D46 MRD-positive external patient).

The Final Risk gap has a different cause. Even when given every input of the protocol rule at $\mathcal{F}_7$, the LLM ensemble assigned the correct group with a balanced accuracy of 0.833—below that of Initial Risk alone (0.906) and of the rule itself (1.000)—whereas the RF reached 0.958 after learning from local cases. Zero-shot LLMs therefore do not apply the local stratification protocol; protocol logic is better encoded explicitly than left to the model.

2.4 Robustness at external hospitals

Applied without retraining to the external cohort (Fig. 4a and Table 3), the trained RSF lost discrimination for EFS (C-index 0.746 internally, 0.608 externally), whereas the LLM ensemble changed little (0.713 and 0.670). For OS, the LLM ensemble reached 0.829 against 0.643 for the RSF (difference +0.187, 95% CI 0.035 to 0.385). Because one source hospital contributed only relapsed patients, we repeated the survival comparison counting only patient pairs from the same hospital; the LLM ensemble remained ahead (EFS 0.702 versus 0.629; OS 0.842 versus 0.691), although with 41 patients

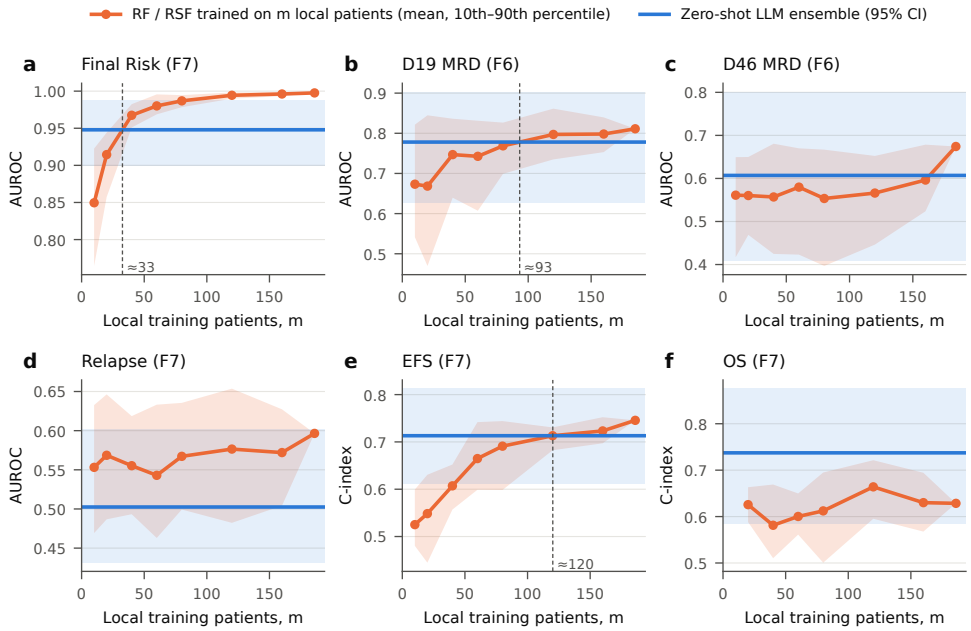


Fig. 3 How many local patients a zero-shot LLM ensemble is worth. Random forests (AUROC) and random survival forests (C-index) were trained on m patients drawn at random from the 2015–17 enrolment years (20 repeats per size; line, mean; band, 10th–90th percentile) and evaluated on the 2018–19 test patients at each target's decision level. The blue line and band show the zero-shot LLM ensemble on the same patients (95% bootstrap CI). Dashed lines mark the training size at which the trained model's mean reaches the LLM ensemble. For D46 MRD and relapse both approaches are close to chance; for OS the trained model stays below the LLM ensemble at every size.

neither within-hospital difference reached significance. Final Risk transferred well for both approaches (AUROC 0.975 and 0.992). The external D19 MRD, D46 MRD and relapse comparisons are not informative: they rest on four positive patients, one positive patient, and a relapse label that almost coincides with hospital of origin, respectively. Kaplan–Meier curves split at the median predicted risk separated EFS internally for both approaches (log-rank $p = 0.010$ for each) and pointed in the same direction externally without reaching significance (LLM ensemble $p = 0.111$; RSF $p = 0.139$; Fig. 4b–e).

2.5 Retrieval-based belief revision and clinical narratives

Revising RF probabilities towards the outcomes of the five most similar training cases (Methods) changed classification AUROC by −0.042 to +0.073 across tasks and splits, apart from a gain of +0.226 for D46 MRD in the 17-patient 2019 test set, which contains only three positives (Supplementary Table S9). For survival, a censoring-aware variant that replaces neighbour outcomes with a similarity-weighted Kaplan–Meier event probability gave small but consistent gains (mean C-index change +0.006 over 70 model, split and endpoint configurations; Wilcoxon $p = 0.001$), concentrated

Table 3 Zero-shot LLM ensemble versus locally trained models at the pre-specified decision level. Internal: 2018–19 enrolment test set (Split 2; models trained on 2015–17). External: two hospitals (models trained on all internal patients). LLM, equal-weight ensemble of four zero-shot LLMs; trained, random forest (classification) or random survival forest (survival). AUROC for Final Risk, D19 MRD, D46 MRD and relapse; Harrell's C-index for EFS and OS. Δ, LLM ensemble minus trained model, with 95% bootstrap CI (2,000 resamples); $^{*}p < 0.05$. Equiv. n, number of local training patients at which the trained model's mean learning curve reaches the LLM ensemble (Fig. 3); n.a., both approaches near chance. †Four positive patients. ‡Relapse status almost coincides with hospital of origin, because one hospital contributed only relapsed patients. Final Risk at $\mathcal{F}_7$ is reproduced by a three-condition protocol rule (Section 2.2). All test sets, balanced accuracies and sensitivity analyses are in Supplementary Table S4.

Target (level)	Test set	LLM	Trained	Δ [95% CI]	Equiv. n
Final Risk ($\mathcal{F}_7$)	Internal	0.948	0.998	-0.050 [-0.097, -0.010]*	$\approx$33
	External	0.975	0.992	-0.017 [-0.067, $+0.000$]	
D19 MRD ($\mathcal{F}_6$)	Internal	0.778	0.811	-0.033 [-0.177, $+0.103$]	$\approx$93
	External	0.440†	0.330†	$+0.110$ [-0.291, $+0.599$]	
D46 MRD ($\mathcal{F}_6$)	Internal	0.607	0.674	-0.067 [-0.270, $+0.133$]	n.a.
	External	not evaluable (1 positive)			
Relapse ($\mathcal{F}_7$)	Internal	0.503	0.597	-0.094 [-0.250, $+0.068$]	n.a.
	External	0.522‡	0.377‡	$+0.145$ [-0.134, $+0.400$]	
EFS ($\mathcal{F}_7$)	Internal	0.713	0.746	-0.033 [-0.125, $+0.060$]	$\approx$120
	External	0.670	0.608	$+0.062$ [-0.089, $+0.224$]	
OS ($\mathcal{F}_7$)	Internal	0.737	0.629	$+0.109$ [$+0.011$, $+0.229$]*	$>$186
	External	0.829	0.643	$+0.187$ [$+0.035$, $+0.385$]*	

in weaker base models. For each test patient, an LLM then produced a short narrative combining the prediction, the decision-tree path and the retrieved cases; a structured physician evaluation of these narratives is under way (Supplementary Table S10).

3 Discussion

Most prediction models in paediatric ALL assume that a hospital's own historical cases are available for training. By aligning six clinical decisions with the timelines along which data are acquired and accumulate, we found that this assumption matters less than expected for several of them. A zero-shot ensemble of four LLMs, using no local data, matched random forests and survival forests trained on roughly 90–120 local patients for D19 MRD and EFS, outperformed them for OS in every evaluable test set, and lost less discrimination when moved to two other hospitals. Trained models were clearly superior only for Final Risk, a target that a three-condition protocol rule reproduces almost exactly.

When a zero-shot LLM suffices. These results help reconcile two apparently conflicting lines of evidence. Prompted LLMs have outperformed conventional models trained on ten examples [33] yet fallen well short of models trained on full datasets [34, 35]. Our learning curves locate the crossover within a single disease: about 30 patients for a protocol-defined target, around 100 for EFS, and beyond the available data for OS, where only 8 to 13 deaths were available for training. The LLMs' advantage is

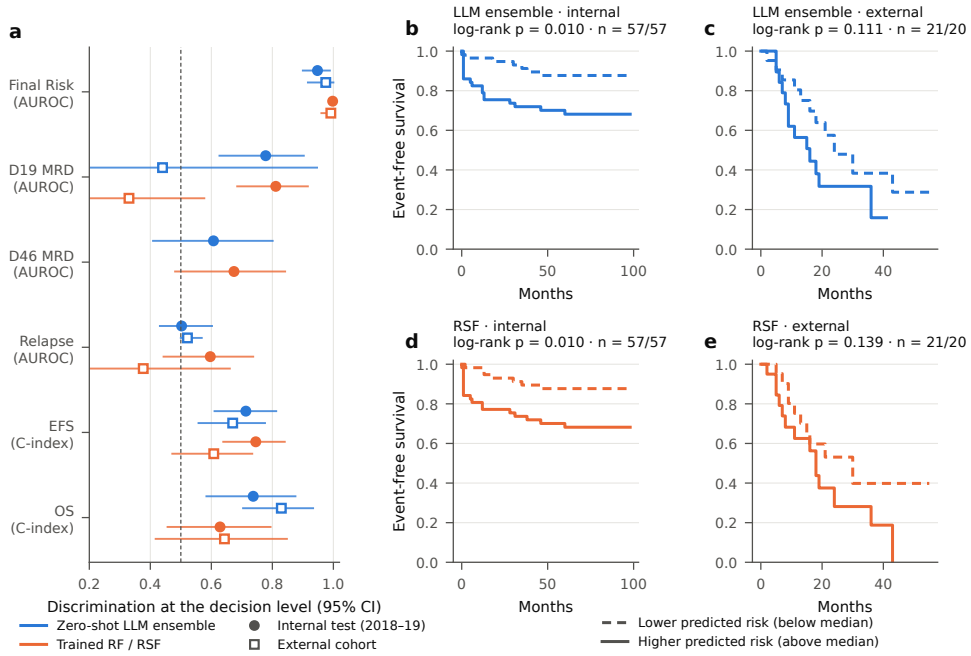


Fig. 4 Internal versus external performance. a, Discrimination at the decision level (95% bootstrap CI) for the zero-shot LLM ensemble and trained random forests or random survival forests, in the internal 2018–19 test set (filled) and the external cohort (open). External D46 MRD (one positive patient) is omitted; external D19 MRD (four positives) and relapse (confounded with hospital of origin) should not be interpreted. **b–e,** Event-free survival by predicted risk (above versus below the median within each cohort) for the LLM ensemble (**b**, **c**) and the random survival forest (**d**, **e**), internally (**b**, **d**) and externally (**c**, **e**); *p*, log-rank test; *n*, patients per group.

largest where local events are scarce and where the case mix shifts, the two situations in which a locally trained model has least to learn from or most to unlearn. In practice, a centre starting a new protocol could use a zero-shot LLM ensemble for survival risk from the first patient, and move to, or combine it with, locally trained models for frequent-event targets once around one hundred cases have accrued.

Where it does not. Given every input of the protocol rule, the LLM ensemble still misassigned Final Risk more often than Initial Risk alone would have. General-purpose LLMs do not know, or do not apply, the thresholds of a local stratification protocol, and decisions defined by such rules should be implemented as explicit logic rather than delegated to a language model. The value of zero-shot LLMs lies instead in outcomes that no rule defines—survival and, to a lesser extent, early treatment response.

MRD and the limits of routine data. Survival discrimination appeared only once D19 and D46 MRD were available, for trained models and LLMs alike, in line with the established role of MRD as the dominant prognostic factor [5, 6]. Relapse and D46 MRD remained near chance at every level, consistent with the failure of computational flow cytometry at diagnosis [24] and the modest discrimination of a large clinical

relapse model [17]. Stronger relapse signal has so far come mainly from single-cell and molecular profiles [20, 22], which our routine feature set does not include; adding such data at the appropriate point of the timeline is a natural next step.

Retrieval-based revision. Revising model probabilities with similar past cases gave small, consistent gains for survival but mixed results for classification. Retrieval helps when the feature space clusters by outcome and neighbours agree, and adds noise when they do not, as for relapse. A practical safeguard would be to revise only when neighbour outcomes agree beyond a calibrated threshold.

Limitations. The internal cohort comes from a single institution, is smaller than the cooperative-group and registry cohorts behind earlier risk models [4, 9, 10], and was analysed retrospectively; prospective deployment, as reported for a clinical language model in adult care [30], remains a stronger standard of validation than our temporal and external evaluation. The external cohort is small (41 patients) and includes a relapse-enriched referral series, so external relapse and MRD results are not interpretable and the within-hospital survival differences did not reach significance; larger, prospectively collected external cohorts are needed to confirm the robustness of the LLM ensemble. Learning curves were estimated on one temporal split, the 2019 test set contains only 17 patients, and no correction was made for multiple comparisons. The LLMs were used zero-shot with fixed prompts; their outputs may change with model versions, and fine-tuning or few-shot prompting might narrow the Final Risk gap. Because each LLM returns a hard label for classification, the ensemble's classification score has only five levels, which limits its AUROC. Finally, physician evaluation of the LLM narratives is in progress and will determine whether they are useful beyond illustrative case studies.

4 Methods

4.1 Datasets and ethics

The internal training cohort comprised $n = 301$ patients diagnosed with ALL at [HOSPITAL NAME] between 2015 and 2019. All patients were treated with risk-stratified chemotherapy protocols consistent with contemporary Chinese paediatric oncology guidelines. Ethics approval was obtained from [ETHICS NUMBER]; all data were de-identified prior to analysis. Each patient record includes structured clinical data (demographics, haematological indices, immunophenotype, 51-marker flow cytometry panel, cytogenetics, MRD values).

The external validation cohort ($n = 41$) was supplied as a single merged file spanning two source hospitals, distinguished by the recorded hospital field: Lianyungang First People's Hospital (20 patients, 2019–2023) and Children's Hospital of Fudan University, Shanghai (21 patients, 2017–2021, a relapse-enriched referral cohort in which all patients relapsed); ethics approval: [ETHICS NUMBERS, external cohort]. We pool the two hospitals and evaluate the external cohort as a single group throughout (Section 4.6). No external data were used in any model training.

4.2 Temporally-structured feature hierarchy

Feature sets are temporally ordered and, except for the $\mathcal{F}_4$ branch, form a strictly nested sequence $\mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \mathcal{F}_3 \subseteq \mathcal{F}_5 \subseteq \mathcal{F}_6 \subseteq \mathcal{F}_7$ ($\mathcal{F}_4$ branches from $\mathcal{F}_3$; Table S3), where each level contains exactly the features available at the corresponding clinical timepoint. Boundaries: $\mathcal{F}_1$ (Day 0): WBC count and age; $\mathcal{F}_2$ (Day 1): adds blast percentage; $\mathcal{F}_3$ (Day 1): adds immunophenotype and 51 flow cytometry markers; $\mathcal{F}_4$ (Day 1): adds non-risk clinical factors; $\mathcal{F}_5$ (Day 5): adds CNS status, Day 5 blast, and Initial Risk; $\mathcal{F}_6$ (Day 7–10): adds Fusion gene, Karyotype, FISH; $\mathcal{F}_7$ (Day 46): adds D19MRD and D46MRD. The complete variable list for every level is given in Table S3. All six prediction targets are evaluated on the same feature hierarchy.

For Phase 2, categorical features were one-hot encoded and numerical features z-score normalised using training-split statistics only:

$$z_{ij}^{(\text{test})} = \frac{x_{ij}^{(\text{test})} - \mu_j^{(\text{train})}}{\sigma_j^{(\text{train})}}, \tag{1}$$

with missing values imputed by training-split medians.

4.3 Phase 1: LLM zero-shot inference

For classification tasks, each patient's features were serialised into a key–value list and presented with a task-specific system prompt requesting a bracketed label (e.g., `[[IR]]`/`[[LR]]`). For survival risk scoring, the same serialised features were presented with a prompt requesting a continuous event probability (0.0–1.0). All experiments used temperature = 0 (greedy decoding); open-weight models were served with vLLM [41]. Because each classification output is a single hard label, classification performance is summarised by balanced accuracy, BA = $\frac{1}{2}$(TPR + TNR), which equals the AUROC of the binary prediction. The four LLMs were combined into an ensemble with equal weights and no tuning. For classification, the ensemble score is the fraction of models predicting the positive class (patients without a valid answer from any model were excluded), and the ensemble label is positive when at least two of the four models agree. For survival, the ensemble score is the mean of the four predicted event probabilities; the mean within-cohort rank of the four scores was used as a sensitivity analysis (Supplementary Table S4). Per-patient outputs were cached; the cohort was scored for every combination of feature level (×7), task (×6), and model (×4).

4.4 Phase 2: Decision tree and random forest

A decision tree (DT; `max_depth`=4, `class_weight`='balanced', `criterion`='gini', `random_state`=42) and a random forest (RF; $n_{\text{estimators}}$ = 100, `class_weight`='balanced') were trained for each feature level and temporal split. Three temporal splits defined by enrolment year threshold $\mathcal{T}_s$ (Eq. 2) ensure strict temporal non-leakage.

$$\mathcal{D}_s^{\text{train}} = \{i : \tau_i \leq \mathcal{T}_s\}, \quad \mathcal{D}_s^{\text{test}} = \{i : \tau_i > \mathcal{T}_s\}. \tag{2}$$

Enrolment year τ_i is taken from the diagnostic smear date; one patient whose date could not be parsed is excluded from all splits. Patients with a missing label for a task are excluded from that task's training and test sets. Primary metric: AUROC; secondary: F1-macro and balanced accuracy.

4.5 Phase 2: Random survival forest

A random survival forest (RSF; [42]; $n_{\text{estimators}} = 100$, `random_state`=42) was trained on the same temporal splits as the classification experiments; a single survival decision tree (survival-tree analogue of the classification DT) was trained under the same protocol as its tree-based counterpart. EFS was defined as time from diagnosis to first of: relapse, induction failure, or death; OS was time to death. Censoring was at last follow-up. Evaluation used Harrell's C-index (Eq. 6), IPCW time-dependent AUC (Eq. 7) at $t^* \in \{12, 24, 36\}$ months, and the log-rank test on Kaplan-Meier curves dichotomised at the median risk score.

4.6 External OOD evaluation

For external validation, all $n = 301$ internal patients served as the training set and external patients served as the test set; no site-specific fine-tuning was performed. DT and RF were trained on the full internal set using the same preprocessing pipeline (Eq. 1). RSF was trained identically. For Phase 1 LLM evaluation, no training was involved; the same zero-shot prompts were applied to external patient feature vectors.

4.7 Decision levels, learning curves and protocol baseline

Before comparing methods, we fixed one decision level per target: $\mathcal{F}_6$ for D19 and D46 MRD, the last level before the Day-19 assessment, and $\mathcal{F}_7$ for Final Risk, relapse, EFS and OS, which are assigned or predicted after induction. All comparisons between the LLM ensemble and trained models use this level and the same test patients. For balanced accuracy, RF probabilities were thresholded at the training-set prevalence. Learning curves were computed on Split 2: for $m \in \{10, 20, 40, 60, 80, 120, 160\}$ and for the full training set, RF or RSF models with the settings above were trained on m patients drawn without replacement from the 2015–17 enrolment years (20 repeats per m) and evaluated on the 2018–19 test patients with a valid LLM prediction. The LLM-equivalent training size is the value of m at which the mean learning curve first reaches the LLM ensemble, obtained by linear interpolation between adjacent sizes. As a reference for Final Risk, a protocol rule assigned IR/HR when Initial Risk was IR or HR, D19 MRD was $\geq 1\%$, or D46 MRD was $\geq 0.01\%$.

4.8 RAG belief revision and LLM narration

For test patient i, the $K = 5$ most similar training cases were retrieved by cosine similarity in the standardised feature space:

$$\mathcal{N}(i) = \underset{j \in \mathcal{D}^{\text{train}}}{\arg\,\text{top-}K} \; \frac{\mathbf{z}_i \cdot \mathbf{z}_j}{\|\mathbf{z}_i\|_2 \|\mathbf{z}_j\|_2}, \quad K = 5. \tag{3}$$

The RAG evidence probability is

$$p_{\text{rag}} = \frac{\sum_{j \in \mathcal{N}(i)} s_j \, y_j}{\sum_{j \in \mathcal{N}(i)} s_j}, \tag{4}$$

where s_j is the cosine similarity to neighbour j and $y_j \in \{0, 1\}$ is the binary task outcome (or event indicator for survival). The revised probability is

$$p_{\text{rev}} = \alpha \, p_0 + (1 - \alpha) \, p_{\text{rag}}, \qquad \alpha = 1 - (1 - \alpha_{\text{base}})\sqrt{\bar{s}}, \tag{5}$$

where $\alpha_{\text{base}} = 0.6$ and $\bar{s} = \frac{1}{K} \sum_{j \in \mathcal{N}(i)} s_j$. For survival, p_0 is the min-max normalised RSF risk score.

After belief revision, an LLM prompt containing the patient's features, the human-readable DT decision path [43], top-6 salient features (ranked by importance$_j \times |z_{ij}|$), and top-5 retrieved case profiles was submitted to Gemini-3.1-Flash-Lite to generate a 3–5 sentence clinical narrative.

4.9 Survival metrics

Harrell's concordance index:

$$\hat{C} = \frac{\sum_{i \neq j} \mathbf{1}[T_i < T_j] \, \mathbf{1}[\delta_i = 1] \, \mathbf{1}[\hat{r}_i > \hat{r}_j]}{\sum_{i \neq j} \mathbf{1}[T_i < T_j] \, \mathbf{1}[\delta_i = 1]}. \tag{6}$$

IPCW time-dependent AUC:

$$\widehat{\text{AUC}}(t^*) = \frac{\sum_{i:\, T_i \leq t^*,\, \delta_i = 1} \sum_{j:\, T_j > t^*} \mathbf{1}[\hat{r}_i > \hat{r}_j] \, \hat{w}_i}{\sum_{i:\, T_i \leq t^*,\, \delta_i = 1} \sum_{j:\, T_j > t^*} \hat{w}_i}, \tag{7}$$

where $\hat{w}_i = 1/\hat{G}(T_i)^2$ and $\hat{G}(\cdot)$ is the Kaplan-Meier censoring estimator [44].

4.10 Statistical analysis

AUROC and balanced accuracy were computed with `scikit-learn` [45], and C-index and IPCW AUC with `scikit-survival` [46]. Confidence intervals are percentile intervals from 2,000 patient-level bootstrap resamples. Differences between the LLM ensemble and trained models were computed on the same resamples, with two-sided p-values given by twice the smaller tail of the bootstrap distribution of the difference. In the external cohort we also computed a within-hospital C-index that counts only patient pairs from the same hospital, with bootstrap resampling stratified by hospital. Kaplan–Meier curves were compared by the log-rank test after splitting patients at the median predicted risk within each cohort. Split 3 contains only 17 test patients (8 IR/HR, and 3 positives each for D19 MRD, D46 MRD and relapse), so its estimates are indicative only. No correction was made for multiple comparisons; all p-values are two-sided.

Data availability

The clinical dataset contains patient health information and cannot be made publicly available due to privacy regulations. De-identified data may be available through a data access agreement with [HOSPITAL NAME]; requests to the corresponding author.

Code availability

Analysis code and LLM prompts will be made available at https://github.com/YanbeiJiang/ALL_prediction_NCS upon acceptance.

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Author contributions

[To be completed]

Competing interests

The authors declare no competing interests.

Appendix A Supplementary tables

Table S1 Characteristics of individual prior outcome-prediction studies, grouped by family and compared with this work (summary by dimension in Table 1). Data: Clin, clinical and laboratory variables; Cyto, cytogenetics or molecular genetics; MRD, minimal residual disease; Flow, flow cytometry; EHR, structured electronic health records; Notes, free-text clinical notes; KG, knowledge graph. Endpoints: * modelled jointly by one model; otherwise separate models or a single endpoint. TL (timeline): ✓ only features available at the time of prediction; × features from different time points combined, or post-prediction variables included; part. partially; nested features organised into timeline-ordered nested levels and evaluated at each decision node. Validation: App. apparent (no hold-out); Rand. random split; CV cross-validation; Temp. temporal split; Ext. independent external cohort; Prosp. prospective deployment; ‡ the set described as independent or external comes from the same source; § the external cohort was not used to evaluate outcome prediction. LLM: ZS zero-shot prompting or generation; FT domain-pretrained or fine-tuned language model; RAG retrieval augmentation.

Study	Population (n)	Model	Data	Endpoints	TL	Validation	LLM
Paediatric ALL: risk-stratification rules and statistical models							
Schultz 2007 [4]	B-ALL (6,238)	Rule-based groups	Clin, Cyto, MRD	EFS	✓	App.	–
Borowitz 2008 [5]	B-ALL (2,143)	Prognostic factors	Clin, MRD	EFS	✓	App.	–
Conter 2010 [6]	BCP-ALL (3,184)	MRD-based rule	MRD	EFS	✓	App.	–
Mao 2020 [7]	ALL (673 + 299)	Cox nomogram	Clin, Cyto, MRD	OS	×	Rand.‡	–
He 2022 [9]	ALL (1,693)	Cox nomogram	Clin, Cyto, MRD	EFS	×	Rand.‡	–
Liu 2022 [11]	T-ALL (105)	Weighted score	Clin, Cyto, MRD	OS, EFS, RFS	✓	App.	–
Chen 2025 [12]	ALL (436)	Logistic regression	Clin, Cyto, MRD	Relapse	✓	App.	–
Paediatric ALL: machine-learning and omics models							
Kang 2010 [18]	B-ALL (207 + 84)	Expression classifiers	Transcriptome, Clin, MRD	RFS; day-29 MRD	part.	CV + Ext.	–
Pan 2017 [13]	ALL (570)	Random forest	Clin	Relapse	×	Rand., CV	–
Good 2018 [20]	BCP-ALL (54)	Single-cell states + Cox	Mass cytometry, Clin	Relapse	✓	Rand.‡	–
Bohannan 2022 [21]	HR B-ALL (156)	Random survival forest	Exome	EFS	✓	Rand. + Ext.§	–
Mehrbakhsh 2024 [14]	ALL (161)	10 ML classifiers	Clin	Death; relapse	✓	Rand.	–
Mosquera Orgueira 2024 [22]	ALL (763 + 426)	Random survival forest	DNA methylation	RFS; OS	✓	Rand. + Ext.	–
Martínez-Rubio 2025 [24]	B-ALL (188)	FlowSOM + ML	Flow	Relapse	✓	Nested CV	–
Song 2025 [15]	Ph-like ALL (226)	Random forest	Clin, MRD	EFS	×	Rand., CV‡	–
Saravanan 2026 [17]	ALL (2,252)	XGBoost	Clin, Cyto, MRD	Relapse	✓	Rand.	–
Zhai 2026 [23]	B-ALL (531)	RNA-seq groups + Cox	Transcriptome, Clin, MRD	EFS, OS	✓	App.	–
Stoiber 2026 [27]	T-ALL (1,309 + 228)	Multi-task network	Clin, genome, transcriptome	Subtype, events, EFS*	✓	CV + Ext.§	–
Language models and retrieval for clinical prediction (adult, non-leukaemia)							
Jiang 2023 [30]	Adult inpatients	Clinical LM (NYUTron)	Notes	5 tasks	✓	Temp. + Prosp.	FT
Renc 2024 [31]	Adult (MIMIC-IV)	EHR transformer (ETHOS)	EHR events	Multiple*	✓	Rand.	ZS
Brown 2025 [35]	Adult (2 sites)	LLM vs boosted trees	EHR	2 tasks	✓	CV, bootstrap	ZS, RAG
Zhu 2026 [33]	Adult (MIMIC, TJH)	LLM vs ML (ClinicRealm)	EHR	3 tasks	✓	Rand.	ZS
Chen 2026 [34]	Adult (MIMIC)	LLM vs ML (ClinicalBench)	EHR codes as text	3 tasks	✓	Rand.	ZS
Chen 2026 [36]	Adult ED (87,112)	LLM vs LightGBM	EHR, Notes	ED revisit	✓	Temp.	ZS, RAG
Datta 2026 [37]	Adult ICU (MIMIC-III)	LLM + ML (KAMELEON)	EHR, Notes, KG	2 tasks*	part.	Rand.	FT, RAG
Zhu 2024 [39]	Adult ICU (MIMIC)	LLM + fusion (EMERGE)	EHR, Notes, KG	2 tasks	✓	Rand.	RAG
Kiermeyer 2026 [32]	Adult cancer (3,522)	LLM features + RSF	Notes, TNM	OS	✓	CV	ZS
This work	Paed. ALL (301 + 41)	LLMs; DT/RF/RSF; RAG	Clin, Flow, Cyto, MRD	6 targets*	nested	Temp. + Ext.	ZS, RAG

Table S2 Feature set design aligned with the clinical data-acquisition timeline. All six prediction targets are evaluated at every feature level; † at $\mathcal{F}_7$, D19MRD and D46MRD are used as *input features*, so only Final Risk and Relapse are predicted. EFS/OS are evaluated at all levels using the same feature vector as classification tasks.

Level	**Timepoint**	**Cumulative features added**	**Class. targets**
$\mathcal{F}_1$	Day 0	WBC count, Age	4
$\mathcal{F}_2$	Day 1	+ Blast%	4
$\mathcal{F}_3$	Day 1	+ Immunophenotype + 51 flow cytometry markers	4
$\mathcal{F}_4$	Day 1	$\mathcal{F}_3$ + non-risk clinical factors	4
$\mathcal{F}_5$	Day 5	$\mathcal{F}_3$ + CNS status + Day 5 blast% + Initial Risk	4
$\mathcal{F}_6$	Day 7–10	$\mathcal{F}_5$ + Fusion gene + Karyotype + FISH	4
$\mathcal{F}_7$	Day 46	$\mathcal{F}_6$ + D19MRD + D46MRD (as inputs)	$2^\dagger$

Survival targets (EFS, OS) evaluated at all levels using the same feature vector.

Table S3 Complete variable specification for each feature level (Supplementary Table). Each level adds the listed variables to its base set; $\mathcal{F}_4$ branches from $\mathcal{F}_3$, while the main clinical chain is $\mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \mathcal{F}_3 \subseteq \mathcal{F}_5 \subseteq \mathcal{F}_6 \subseteq \mathcal{F}_7$.

Level	Variables added (base set)
$\mathcal{F}_1$	WBC count; Age
$\mathcal{F}_2$	+ Blast%
$\mathcal{F}_3$	+ Immunophenotype (B/T lineage); 42 CD markers (CD1a, CD2, CD3, cCD3, CD4, CD5, CD7, CD8, CD9, CD10, CD11b, CD11c, CD13, CD14, CD15, CD19, CD20, CD22, cCD22, CD24, CD33, CD34, CD38, CD41, CD44, CD45, CD58, CD61, CD64, CD65, CD66c, CD71, CD73, cCD79a, CD86, CD90, CD97, CD99, CD117, CD123, CD133, CD200); 9 additional markers (GPA, cMPO, NG2, TdT, cμ, sIgM, TCR$\alpha\beta$, TCR$\gamma\delta$, HLA-DR)
$\mathcal{F}_4$	(base $\mathcal{F}_3$) + non-risk clinical factors: chief-complaint category, pregnancy number, birth order, term/preterm status, delivery mode, birth weight, feeding type, admission weight, liver span, spleen span, haemoglobin, platelet count
$\mathcal{F}_5$	(base $\mathcal{F}_3$) + CNS status (CNSL); Day-5 peripheral blast%; Initial Risk
$\mathcal{F}_6$	(base $\mathcal{F}_5$) + Fusion gene; Karyotype; FISH
$\mathcal{F}_7$	(base $\mathcal{F}_6$) + D19MRD; D46MRD (as input features)

Table S4 Zero-shot LLM ensemble versus trained models at the decision level in every test set. AUROC (classification) or Harrell's C-index (survival) with 95% bootstrap CI (2,000 resamples), followed by balanced accuracy (BA; LLM ensemble: majority of the four models; trained: threshold at the training-set prevalence). Trained, random forest (classification) or random survival forest (survival). Notes: for Final Risk, BA of the three-condition protocol rule and of Initial Risk alone; for survival, C-index of the rank-mean ensemble (sensitivity analysis) and, in the external cohort, within-hospital C-index (LLM ensemble vs trained). (n, patients and positives or events.)

Test set	LLM ensemble	Trained	BA LLM	BA trained	Notes
Final Risk (F7)					
Split 1 (2017–19) (n=200, 100)	0.950 [0.919, 0.979]	0.996 [0.990, 0.999]	0.850	0.950	rule 1.000; Initial Risk 0.895
Split 2 (2018–19) (n=113, 48)	0.948 [0.900, 0.988]	0.998 [0.992, 1.000]	0.833	0.958	rule 1.000; Initial Risk 0.906
Split 3 (2019) (n=17, 8)	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000	1.000	rule 1.000; Initial Risk 0.813
External (n=29, 20)	0.975 [0.917, 1.000]	0.992 [0.961, 1.000]	0.900	0.919	rule 0.975; Initial Risk 0.900
D19 MRD (F6)					
Split 1 (2017–19) (n=200, 31)	0.668 [0.563, 0.773]	0.684 [0.562, 0.797]	0.647	0.642	
Split 2 (2018–19) (n=113, 13)	0.778 [0.627, 0.903]	0.811 [0.685, 0.916]	0.720	0.725	
Split 3 (2019) (n=17, 3)	0.500 [0.167, 0.844]	0.631 [0.333, 0.933]	0.452	0.524	
External (n=29, 4)	0.440 [0.107, 0.946]	0.330 [0.120, 0.577]	0.365	0.345	
D46 MRD (F6)					
Split 1 (2017–19) (n=200, 21)	0.539 [0.413, 0.664]	0.586 [0.465, 0.706]	0.594	0.559	
Split 2 (2018–19) (n=113, 11)	0.607 [0.409, 0.801]	0.674 [0.481, 0.842]	0.616	0.663	
Split 3 (2019) (n=17, 3)	0.631 [0.369, 0.867]	0.429 [0.067, 0.867]	0.488	0.452	
External	not evaluable				
Relapse (F7)					
Split 1 (2017–19) (n=200, 28)	0.572 [0.487, 0.658]	0.590 [0.487, 0.688]	0.572	0.525	
Split 2 (2018–19) (n=113, 17)	0.503 [0.431, 0.602]	0.597 [0.443, 0.737]	0.517	0.567	
Split 3 (2019) (n=17, 3)	0.357 [0.231, 0.464]	0.429 [0.000, 0.813]	0.393	0.619	
External (n=29, 23)	0.522 [0.500, 0.569]	0.377 [0.116, 0.661]	0.522	0.384	
EFS (F7)					
Split 1 (2017–19) (n=201, 43)	0.672 [0.592, 0.752]	0.684 [0.605, 0.766]	–	–	rank mean 0.680
Split 2 (2018–19) (n=114, 25)	0.713 [0.611, 0.813]	0.746 [0.639, 0.841]	–	–	rank mean 0.705
Split 3 (2019) (n=17, 5)	0.684 [0.437, 0.894]	0.691 [0.327, 0.982]	–	–	rank mean 0.684
External (n=41, 24)	0.670 [0.558, 0.776]	0.608 [0.472, 0.734]	–	–	rank mean 0.688; within hospital 0.702 vs 0.629
OS (F7)					
Split 1 (2017–19) (n=201, 15)	0.712 [0.553, 0.852]	0.607 [0.447, 0.757]	–	–	rank mean 0.722
Split 2 (2018–19) (n=114, 10)	0.737 [0.584, 0.876]	0.629 [0.457, 0.794]	–	–	rank mean 0.737
Split 3 (2019)	not evaluable				
External (n=41, 7)	0.829 [0.704, 0.934]	0.643 [0.418, 0.848]	–	–	rank mean 0.788; within hospital 0.842 vs 0.691

Table S5 Phase 1 LLM zero-shot performance across all seven clinical feature levels and six prediction targets (internal cohort; classification $n = 298$ patients with all four labels, survival $n = 301$). Each cell is the balanced accuracy (BA; classification targets) or Harrell's concordance index (survival targets) for one LLM at one feature level; **bold** marks each model's best feature level for that target (column maximum). BA is computed from the parsed categorical prediction (equivalently, the AUROC of the binary prediction). — indicates a target not evaluated at that level: D19/D46MRD are used as *input* features at $\mathcal{F}_7$ (see Table S2).

	Model					Model			
Lv.	Gem.	Gma.	Int.	Qw.	**Lv.**	Gem.	Gma.	Int.	Qw.
Final Risk (BA)					*Relapse (BA)*				
$\mathcal{F}_1$	0.788	0.785	0.729	0.792	$\mathcal{F}_1$	0.514	0.572	0.566	0.530
$\mathcal{F}_2$	0.782	0.796	0.539	0.789	$\mathcal{F}_2$	0.557	0.564	0.519	0.526
$\mathcal{F}_3$	0.812	0.842	0.583	0.831	$\mathcal{F}_3$	0.577	0.572	0.503	0.503
$\mathcal{F}_4$	0.826	0.835	0.591	0.837	$\mathcal{F}_4$	0.575	0.555	0.500	0.552
$\mathcal{F}_5$	0.791	0.854	0.823	0.845	$\mathcal{F}_5$	0.596	0.558	0.554	0.504
$\mathcal{F}_6$	0.834	**0.891**	**0.848**	**0.882**	$\mathcal{F}_6$	**0.599**	0.552	**0.643**	0.546
$\mathcal{F}_7$	**0.849**	0.788	0.716	0.856	$\mathcal{F}_7$	0.567	**0.586**	0.540	**0.557**
D19 MRD (BA)					*EFS (C-index)*				
$\mathcal{F}_1$	0.571	0.543	0.564	0.531	$\mathcal{F}_1$	0.577	0.591	0.584	0.573
$\mathcal{F}_2$	0.543	0.555	0.479	0.521	$\mathcal{F}_2$	0.607	0.598	0.522	0.561
$\mathcal{F}_3$	0.631	0.599	0.484	0.557	$\mathcal{F}_3$	0.587	0.614	0.545	0.537
$\mathcal{F}_4$	0.583	0.597	0.488	0.520	$\mathcal{F}_4$	0.598	0.614	0.526	0.561
$\mathcal{F}_5$	0.603	0.620	**0.567**	**0.605**	$\mathcal{F}_5$	0.624	0.583	0.604	0.542
$\mathcal{F}_6$	**0.651**	**0.652**	0.548	0.581	$\mathcal{F}_6$	0.643	0.630	0.621	0.581
$\mathcal{F}_7$	—	—	—	—	$\mathcal{F}_7$	**0.687**	**0.716**	**0.642**	**0.662**
D46 MRD (BA)					*OS (C-index)*				
$\mathcal{F}_1$	0.505	0.486	**0.542**	0.501	$\mathcal{F}_1$	0.689	0.648	0.652	0.661
$\mathcal{F}_2$	0.492	0.475	0.501	0.495	$\mathcal{F}_2$	0.647	0.702	0.550	0.634
$\mathcal{F}_3$	0.502	0.509	0.503	0.500	$\mathcal{F}_3$	0.682	0.678	0.537	0.634
$\mathcal{F}_4$	0.505	0.539	0.505	0.533	$\mathcal{F}_4$	0.687	0.691	0.497	0.660
$\mathcal{F}_5$	0.549	**0.577**	0.518	**0.552**	$\mathcal{F}_5$	0.701	0.704	0.734	0.619
$\mathcal{F}_6$	**0.610**	0.539	0.523	0.510	$\mathcal{F}_6$	0.667	0.693	0.755	0.687
$\mathcal{F}_7$	—	—	—	—	$\mathcal{F}_7$	**0.725**	**0.718**	**0.764**	**0.718**

Table S6 Phase 1 LLM zero-shot on the pooled external cohort, all feature levels (Gm/Gma/Iv/Qw = Gemini/Gemma/InternVL/Qwen). The two source hospitals are pooled into a single external cohort (n=41 for survival; classification targets use the n=29 subset with available labels). Classification: BA (* = accuracy; D46MRD has only one positive class after pooling, so accuracy is reported instead). Survival (EFS, OS): C-index. **Bold**: best feature level per column within each task block (ties all bolded). —: not evaluable ($\mathcal{F}_7$ not defined for D19/D46 MRD). External pooled cohort; exploratory, indicative only.

Lv.	Gm	Gma	Iv	Qw
Final Risk (BA)				
$\mathcal{F}_1$	0.739	0.794	0.714	0.794
$\mathcal{F}_2$	0.794	0.819	0.450	0.794
$\mathcal{F}_3$	0.708	0.819	0.531	0.794
$\mathcal{F}_4$	0.844	0.819	0.697	0.794
$\mathcal{F}_5$	0.653	**0.894**	0.536	0.739
$\mathcal{F}_6$	0.708	0.844	0.733	0.819
$\mathcal{F}_7$	**0.925**	0.775	**0.750**	**0.850**
D19 MRD (BA)				
$\mathcal{F}_1$	0.425	0.425	0.490	0.380
$\mathcal{F}_2$	0.530	0.405	0.500	**0.550**
$\mathcal{F}_3$	0.425	0.405	0.500	0.340
$\mathcal{F}_4$	**0.550**	0.405	0.500	0.445
$\mathcal{F}_5$	0.425	**0.465**	0.395	0.445
$\mathcal{F}_6$	0.425	0.405	**0.515**	0.545
$\mathcal{F}_7$	—	—	—	—
D46 MRD (accuracy; single positive)				
$\mathcal{F}_1$	**0.690***	0.586*	**0.586***	0.621*
$\mathcal{F}_2$	0.586*	0.552*	0.035*	0.655*
$\mathcal{F}_3$	0.586*	0.586*	0.035*	0.690*
$\mathcal{F}_4$	0.552*	0.586*	0.035*	**0.724***
$\mathcal{F}_5$	0.586*	**0.621***	0.138*	0.690*
$\mathcal{F}_6$	0.621*	0.586*	0.207*	0.655*
$\mathcal{F}_7$	—	—	—	—
Relapse (BA)				
$\mathcal{F}_1$	0.587	0.634	**0.699**	0.482
$\mathcal{F}_2$	0.504	0.612	0.500	0.482
$\mathcal{F}_3$	0.674	0.551	0.562	0.544
$\mathcal{F}_4$	0.609	0.551	0.685	0.565
$\mathcal{F}_5$	**0.717**	0.612	0.620	0.587
$\mathcal{F}_6$	0.634	**0.656**	0.536	**0.609**
$\mathcal{F}_7$	0.500	0.522	0.500	0.522
EFS (C-index)				
$\mathcal{F}_1$	0.657	**0.669**	0.627	0.634
$\mathcal{F}_2$	**0.715**	0.659	0.551	**0.645**
$\mathcal{F}_3$	0.635	0.590	0.615	0.638
$\mathcal{F}_4$	0.634	0.590	0.615	0.638
$\mathcal{F}_5$	0.617	0.656	0.639	0.542
$\mathcal{F}_6$	0.657	0.651	**0.709**	0.567
$\mathcal{F}_7$	0.656	0.664	0.702	0.591
OS (C-index)				
$\mathcal{F}_1$	0.808	**0.821**	0.802	0.714
$\mathcal{F}_2$	0.804	0.792	0.633	0.744
$\mathcal{F}_3$	0.746	0.780	0.710	0.548
$\mathcal{F}_4$	0.760	0.780	0.710	0.548
$\mathcal{F}_5$	0.762	0.808	0.816	0.694
$\mathcal{F}_6$	**0.810**	0.603	0.814	**0.806**
$\mathcal{F}_7$	0.802	0.615	**0.859**	0.665

Table S7 Phase 2 internal results across all feature levels and the three temporal splits (Split-2 values at the decision level are compared with the LLM ensemble in Table 3). Classification: DT / RF AUROC; survival (EFS, OS): survival-tree (DT) / RSF C-index. **Bold** marks the best feature level in each column. Splits by enrolment year: S1 train ≤16/test 17–19; S2 ≤17/18–19; S3 ≤18/19. D19/D46MRD exclude patients without a recorded MRD value. —: not evaluable (D19/D46MRD are inputs at $\mathcal{F}_7$; no OS events in the 2019 test set).

	Split 1		**Split 2**		**Split 3**	
Lv.	DT	RF	DT	RF	DT	RF
Final Risk (DT / RF AUROC)						
$\mathcal{F}_1$	0.759	0.842	0.810	0.871	0.799	0.806
$\mathcal{F}_2$	0.759	0.820	0.810	0.830	0.778	0.847
$\mathcal{F}_3$	0.711	0.849	0.782	0.872	0.771	0.743
$\mathcal{F}_4$	0.711	0.852	0.757	0.847	0.771	0.799
$\mathcal{F}_5$	**0.893**	0.897	0.909	0.943	0.736	0.847
$\mathcal{F}_6$	0.893	0.922	0.911	0.951	0.736	0.882
$\mathcal{F}_7$	0.891	**0.996**	**0.949**	**0.998**	**0.979**	**1.000**
D19 MRD (DT / RF AUROC)						
$\mathcal{F}_1$	0.537	0.523	0.485	0.420	0.393	0.595
$\mathcal{F}_2$	0.503	0.509	0.498	0.505	0.464	0.762
$\mathcal{F}_3$	0.557	0.657	0.686	0.791	**0.893**	0.810
$\mathcal{F}_4$	0.557	0.672	**0.705**	0.799	0.893	0.810
$\mathcal{F}_5$	**0.560**	0.631	0.686	0.740	0.750	**0.845**
$\mathcal{F}_6$	0.557	**0.684**	0.686	**0.811**	0.738	0.631
$\mathcal{F}_7$	—	—	—	—	—	—
D46 MRD (DT / RF AUROC)						
$\mathcal{F}_1$	0.362	0.503	0.456	0.447	**0.464**	0.631
$\mathcal{F}_2$	0.518	0.428	0.456	0.527	0.321	**0.655**
$\mathcal{F}_3$	**0.545**	0.555	0.521	0.570	0.369	0.321
$\mathcal{F}_4$	0.531	0.554	0.480	0.595	0.369	0.298
$\mathcal{F}_5$	0.545	**0.601**	**0.549**	0.636	0.369	0.345
$\mathcal{F}_6$	0.545	0.586	0.521	**0.674**	0.369	0.429
$\mathcal{F}_7$	—	—	—	—	—	—
Relapse (DT / RF AUROC)						
$\mathcal{F}_1$	0.480	0.439	**0.529**	0.465	0.274	0.476
$\mathcal{F}_2$	**0.550**	0.546	0.483	0.526	**0.500**	0.357
$\mathcal{F}_3$	0.528	0.483	0.411	0.485	0.286	0.274
$\mathcal{F}_4$	0.540	0.564	0.411	0.471	0.286	0.036
$\mathcal{F}_5$	0.527	0.517	0.411	0.624	0.286	0.286
$\mathcal{F}_6$	0.528	0.547	0.411	**0.630**	0.488	**0.500**
$\mathcal{F}_7$	0.529	**0.591**	0.411	0.597	0.488	0.429
EFS (DT / RSF C-index)						
$\mathcal{F}_1$	0.468	0.532	0.443	0.610	**0.691**	**0.706**
$\mathcal{F}_2$	0.482	0.516	0.501	0.597	0.529	0.559
$\mathcal{F}_3$	0.527	0.542	0.477	0.552	0.390	0.500
$\mathcal{F}_4$	0.546	0.520	0.482	0.533	0.426	0.485
$\mathcal{F}_5$	0.494	0.511	0.490	0.572	0.390	0.588
$\mathcal{F}_6$	0.497	0.560	0.485	0.592	0.390	0.662
$\mathcal{F}_7$	**0.674**	**0.684**	**0.718**	**0.746**	0.640	0.691
OS (DT / RSF C-index)						
$\mathcal{F}_1$	0.541	**0.688**	0.531	0.497	—	—
$\mathcal{F}_2$	0.467	0.651	0.541	0.540	—	—
$\mathcal{F}_3$	0.512	0.556	0.440	0.515	—	—
$\mathcal{F}_4$	0.498	0.590	0.472	0.529	—	—
$\mathcal{F}_5$	0.492	0.586	0.445	0.622	—	—
$\mathcal{F}_6$	**0.562**	0.621	**0.562**	0.620	—	—
$\mathcal{F}_7$	0.562	0.607	0.505	**0.628**	—	—

Table S8 Phase 2 out-of-distribution results on the pooled external cohort, across all feature levels (train = all 301 internal patients; test = the two source hospitals pooled into a single external cohort, n=41; D19MRD and D46MRD use the 29 and 31 patients with a recorded value). Classification: DT / RF AUROC; * = accuracy (D46MRD has only one positive case after pooling, so accuracy is reported instead). Survival (EFS, OS): survival-tree (DT) / random-survival-forest (RSF) C-index. **Bold** marks the best feature level in each column. —: not evaluable (D19/D46MRD are inputs at $\mathcal{F}_7$). Pooled external cohort; small, indicative only.

Lv.	DT	RF
Final Risk (DT / RF)		
$\mathcal{F}_1$	0.805	0.878
$\mathcal{F}_2$	0.827	0.863
$\mathcal{F}_3$	0.779	0.843
$\mathcal{F}_4$	0.778	0.750
$\mathcal{F}_5$	0.912	0.938
$\mathcal{F}_6$	0.912	0.934
$\mathcal{F}_7$	**0.968**	**0.991**
D19 MRD (DT / RF)		
$\mathcal{F}_1$	**0.630**	0.590
$\mathcal{F}_2$	0.505	**0.650**
$\mathcal{F}_3$	0.385	0.265
$\mathcal{F}_4$	0.385	0.285
$\mathcal{F}_5$	0.385	0.290
$\mathcal{F}_6$	0.385	0.330
$\mathcal{F}_7$	—	—
D46 MRD (DT / RF)		
$\mathcal{F}_1$	**0.742***	**0.968***
$\mathcal{F}_2$	0.581*	0.968*
$\mathcal{F}_3$	0.323*	0.968*
$\mathcal{F}_4$	0.290*	0.968*
$\mathcal{F}_5$	0.323*	0.968*
$\mathcal{F}_6$	0.323*	0.968*
$\mathcal{F}_7$	—	—
Relapse (DT / RF)		
$\mathcal{F}_1$	**0.583**	**0.528**
$\mathcal{F}_2$	0.397	0.494
$\mathcal{F}_3$	0.333	0.300
$\mathcal{F}_4$	0.525	0.272
$\mathcal{F}_5$	0.333	0.377
$\mathcal{F}_6$	0.219	0.267
$\mathcal{F}_7$	0.250	0.383
EFS (DT / RSF C-index)		
$\mathcal{F}_1$	**0.603**	0.567
$\mathcal{F}_2$	0.552	0.572
$\mathcal{F}_3$	0.507	0.434
$\mathcal{F}_4$	0.509	0.475
$\mathcal{F}_5$	0.419	0.444
$\mathcal{F}_6$	0.419	0.439
$\mathcal{F}_7$	0.490	**0.608**
OS (DT / RSF C-index)		
$\mathcal{F}_1$	0.619	**0.718**
$\mathcal{F}_2$	0.657	0.718
$\mathcal{F}_3$	0.571	0.468
$\mathcal{F}_4$	0.702	0.512
$\mathcal{F}_5$	0.647	0.556
$\mathcal{F}_6$	0.647	0.512
$\mathcal{F}_7$	**0.710**	0.643

Table S9 Cosine RAG belief revision: RF AUROC before (p_{RF}) and after (p_{rev}) belief revision ($K = 5$, $\alpha_{\mathrm{base}} = 0.6$), best experiment per task. Internal results are reported per temporal split; external results use the pooled external cohort, as the per-site external cohorts are too small or single-class for this analysis. For the survival endpoints (EFS, OS) the values are Harrell's C-index: base is the RSF risk score and the revision replaces the similarity-weighted binary-outcome mean by a similarity-weighted Kaplan–Meier neighbourhood event probability at 24 months (experiment P2-Exp8, the MRD-augmented feature set). **Bold**: largest gain (Split 3 has 17 test patients, 3 of them D46MRD-positive; indicative only). † insufficient positives; not computed. —: too few events (OS, 2019 test split).

Task	**Split**	p_{RF}	p_{rev}	Δ
Internal (temporal splits)				
Final Risk	Split 1	0.996	0.998	+0.002
Final Risk	Split 2	0.998	0.999	+0.002
Final Risk	Split 3	1.000	1.000	+0.000
D19MRD	Split 1	0.657	0.655	−0.003
D19MRD	Split 2	0.791	0.787	−0.004
D19MRD	Split 3	0.810	0.833	+0.024
D46MRD	Split 1	0.428	0.408	−0.020
D46MRD	Split 2	0.527	0.600	+0.073
D46MRD	Split 3	0.655	**0.881**	**+0.226**
Relapse	Split 1	0.547	0.546	−0.001
Relapse	Split 2	0.630	0.588	−0.042
Relapse	Split 3	0.500	0.476	−0.024
Survival endpoints (C-index; RSF base vs KM revision)				
EFS	Split 1	0.684	0.704	+0.020
EFS	Split 2	0.746	0.745	−0.001
EFS	Split 3	0.691	0.706	+0.015
OS	Split 1	0.607	0.617	+0.010
OS	Split 2	0.629	0.634	+0.005
OS	Split 3		—	
External (pooled cohort)				
Final Risk	—	0.991	1.000	+0.009
D19MRD	—	0.265	0.230	−0.035
D46MRD	—		† ($n_{\mathrm{pos}} = 1$)	
Relapse	—	0.267	0.283	+0.016
Survival endpoints (C-index; RSF base vs KM revision)				
EFS	—	0.608	0.613	+0.005
OS	—	0.643	0.659	+0.016

Table S10 Planned physician evaluation of LLM clinical narratives. Each narrative will be assessed on four dimensions by two independent clinicians using a 5-point Likert scale. Results to be filled following structured review.

Case ID	Factual accuracy (1–5)	Retrieval relevance (1–5)	Clinical plausibility (1–5)	Decision helpfulness (1–5)	Overall (1–5)
Case 001	—	—	—	—	—
Case 002	—	—	—	—	—
Case 003	—	—	—	—	—
⋮	⋮	⋮	⋮	⋮	⋮
Case 017	—	—	—	—	—
Mean (SD)	—	—	—	—	—
Inter-rater κ	—	—	—	—	—

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